PART II: PATTERNS

Topic 14: Comparing Multiple Time Series

The Core Concept

Comparing multiple time series means placing two or more different groups on the **same graph over the same time period**. This allows us to determine whether changes are **local** (unique to one unit) or **systemic** (affecting everyone).

Why It Matters: Benchmarking and Context

If use-of-force incidents spike in one precinct, is it because of local leadership issues — or a city-wide factor like a heat wave, a major event, or a change in state law?

- **If both series move together:** The cause is likely external and shared (weather, city-wide policy, a protest).
- **If one series moves while the other stays flat:** The cause is likely internal and localized (a squad's tactics, a new commander, a neighborhood-specific problem).

The Example

Note: All data in this tutorial series are synthetic, created for teaching purposes only. They do not represent real-world incidents or any specific jurisdiction.

Year	Period	Precinct North (UoF)	Precinct South (UoF)	North Arrests
2021	Jan–Mar	28, 26, 30	25, 24, 26	~500 avg
	Apr–Jun	32, 35, 40	24, 25, 23	
	Jul–Sep	42, 38, 34	26, 25, 24	
	Oct–Dec	30, 28, 25	25, 26, 24	
2022	Jan–Mar	22, 20, 24	24, 26, 25	~450 avg
	Apr–Jun	26, 28, 32	23, 25, 24	
	Jul–Sep	35, 30, 28	25, 26, 24	
	Oct–Dec	24, 22, 18	24, 25, 23	
2023	Jan–Mar	12, 10, 14	25, 22, 28	~410 avg
	Apr–Jun	15, 18, 22	24, 26, 24	
	Jul	25	27	700
	Aug	**(Missing)**	23	680
	Sep	19	22	500
	Oct	16	**85 ★ Outlier**	420
	Nov–Dec	14, 11	24, 26	~370 avg

In October 2023:

- Precinct North: 16 incidents (stable, near its seasonal average)
- Precinct South: 85 incidents (a massive spike)